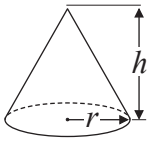
15 A cone has a volume of 98 cm^3.
The radius of the cone is 5.13 cm.

Volume of cone = $\frac{1}{3}\pi r^2 h$



(a) Work out an estimate for the height of the cone.

$V = 98 \approx 100$, $r = 5.13 \approx 5$, $\pi \approx 3$.

$$h = \frac{3V}{\pi r^2} \approx \frac{3 \times 100}{3 \times 5^2}$$

$h \approx 300 \div 75 = 4$ cm

1 mark
Rearrange the volume formula

1 mark
Use rounded values

1 mark
A consistent estimate

4 cm

(3)

John uses a calculator to work out the height of the cone to 2 decimal places.

(b) Will your estimate be more than John's answer or less than John's answer?
Give reasons for your answer.

More. The numerator 3V was rounded up.

The denominator πr^2 was rounded down,

so both changes make the estimated height larger.

1 mark
Both rounding effects explained

(1)

(Total for Question 15 is 4 marks)

16 n is an integer greater than 1

Prove algebraically that $n^2 - 2 - (n - 2)^2$ is always an even number.

(Total for Question 16 is 4 marks)

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